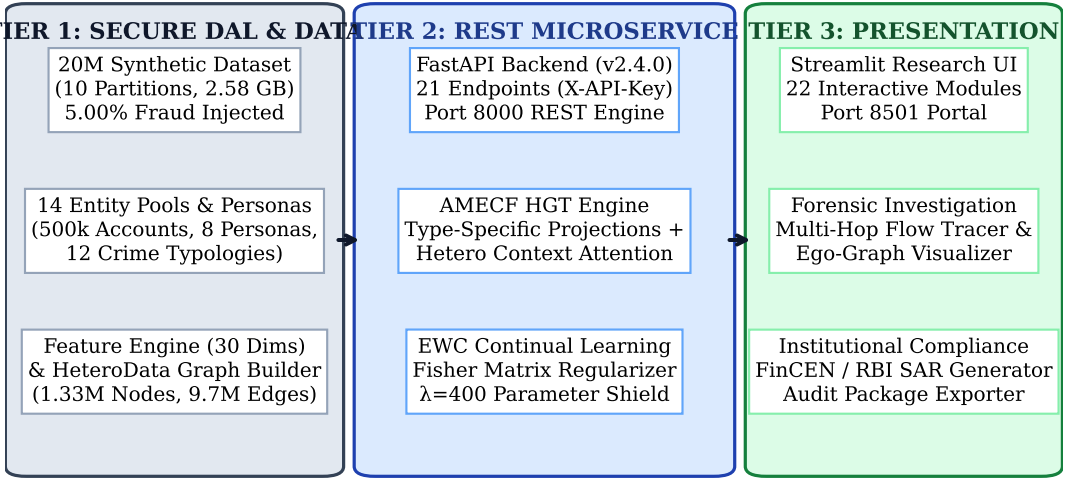
TIER 1: SECURE DAL & DATA
20M Synthetic Dataset
(10 Partitions, 2.58 GB)
5.00% Fraud Injected
14 Entity Pools & Personas
(500k Accounts, 8 Personas,
12 Crime Typologies)
Feature Engine (30 Dims)
& HeteroData Graph Builder
(1.33M Nodes, 9.7M Edges)
TIER 2: REST MICROSERVICE
FastAPI Backend (v2.4.0)
21 Endpoints (X-API-Key)
Port 8000 REST Engine
AMECF HGT Engine
Type-Specific Projections +
Hetero Context Attention
EWC Continual Learning
Fisher Matrix Regularizer
λ=400 Parameter Shield
TIER 3: PRESENTATION
Streamlit Research UI
22 Interactive Modules
Port 8501 Portal
Forensic Investigation
Multi-Hop Flow Tracer &
Ego-Graph Visualizer
Institutional Compliance
FinCEN / RBI SAR Generator
Audit Package Exporter